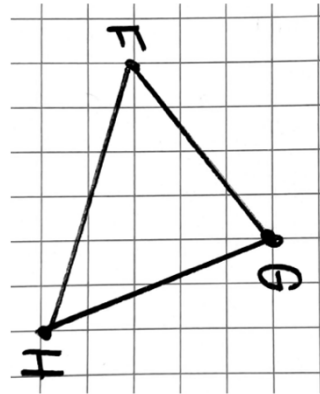
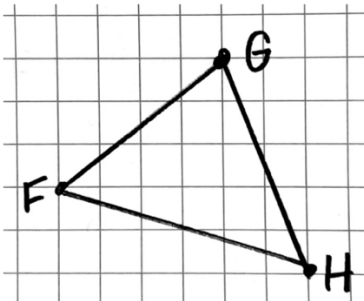


Tamika drew $\triangle FGH$ on graph paper. She then rotated the graph paper and moved it to the right.



1. Explain what method(s) can be used to determine if each side of $\triangle FGH$ has the same length as the triangle that has been rotated and moved.
Patty paper can be used to copy each side of $\triangle FGH$ or a ruler can be used to measure each side of $\triangle FGH$.
2. Explain what method(s) can be used to determine if each angle of $\triangle FGH$ has the same measure as the triangle that has been rotated and moved.
Patty paper can be used to copy each angle of $\triangle FGH$ or a protractor can be used to measure each angle of $\triangle FGH$.
3. Explain why showing that the side lengths and the angles have the same measure shows that the transformations used can be classified as isometry.

Because angle measures and side lengths are preserved.

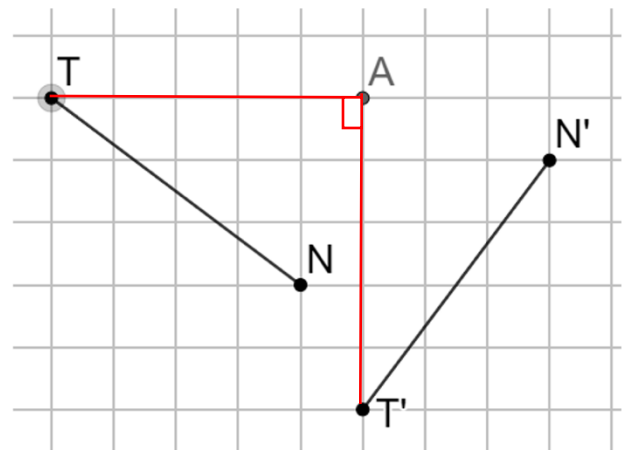
Line segments TN , $T'N'$, and point A are shown on the grid.

4. Draw $\angle TAT'$. Label the angle measure.
 90°
5. Describe the transformation applied to $\overline{TN}$ that resulted in $\overline{T'N'}$.

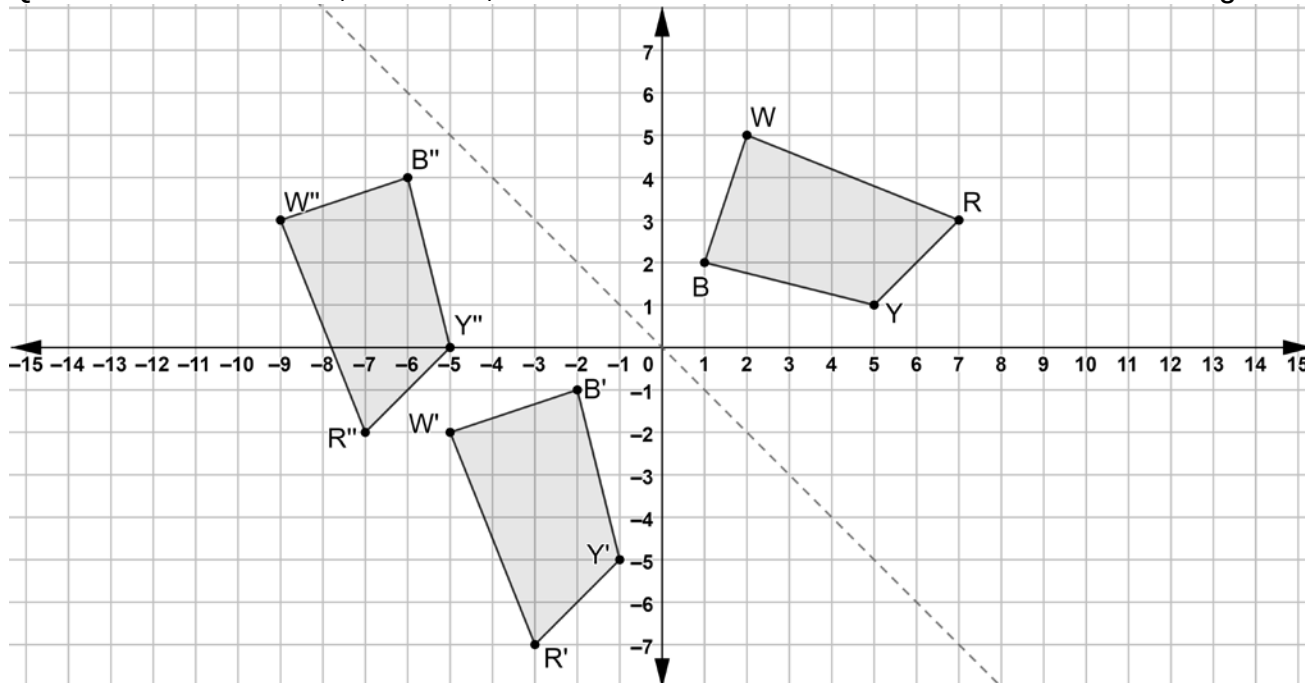
Rotate 90° counterclockwise about Point A .

6. Complete the statement.

There is a(n) isometry between the set of points on $\overline{TN}$ and the set of points on $\overline{T'N'}$, therefore all of the points on $\overline{TN}$ have to have a matching point on $\overline{T'N'}$.



Quadrilaterals $BWRY$, $B'W'R'Y'$, and $B''W''R''Y''$ are shown on the coordinate grid.



7. Describe what movements should be done with a patty paper copy of quadrilateral $BWRY$ so that the copy of $BWRY$ lays on top of $B'W'R'Y'$.

Reflection over the line $y = -x$.

8. Describe what movements should be done with a patty paper copy of quadrilateral $B'W'R'Y'$ so that the copy of $B'W'R'Y'$ lays on top of $B''W''R''Y''$.

Translation left 4 units, up 5 units.

9. Complete the statements.
The transformations used in both sequences

☒ are
☐ are not

examples of isometry. All of the points of

☐ $B'W'R'Y'$
☐ $B''W''R''Y''$
☒ $B'W'R'Y'$ and $B''W''R''Y''$

can be mapped to all of the points of $BWRY$.

10. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $BWRY$ resulted in $BWRY \cong B''W''R''Y''$.

In both sequences of transformations only rigid motions were used to map $BWRY$ to $B'W'R'Y'$ to $B''W''R''Y''$ so the figures are congruent by the definition of congruence in terms of rigid motions.